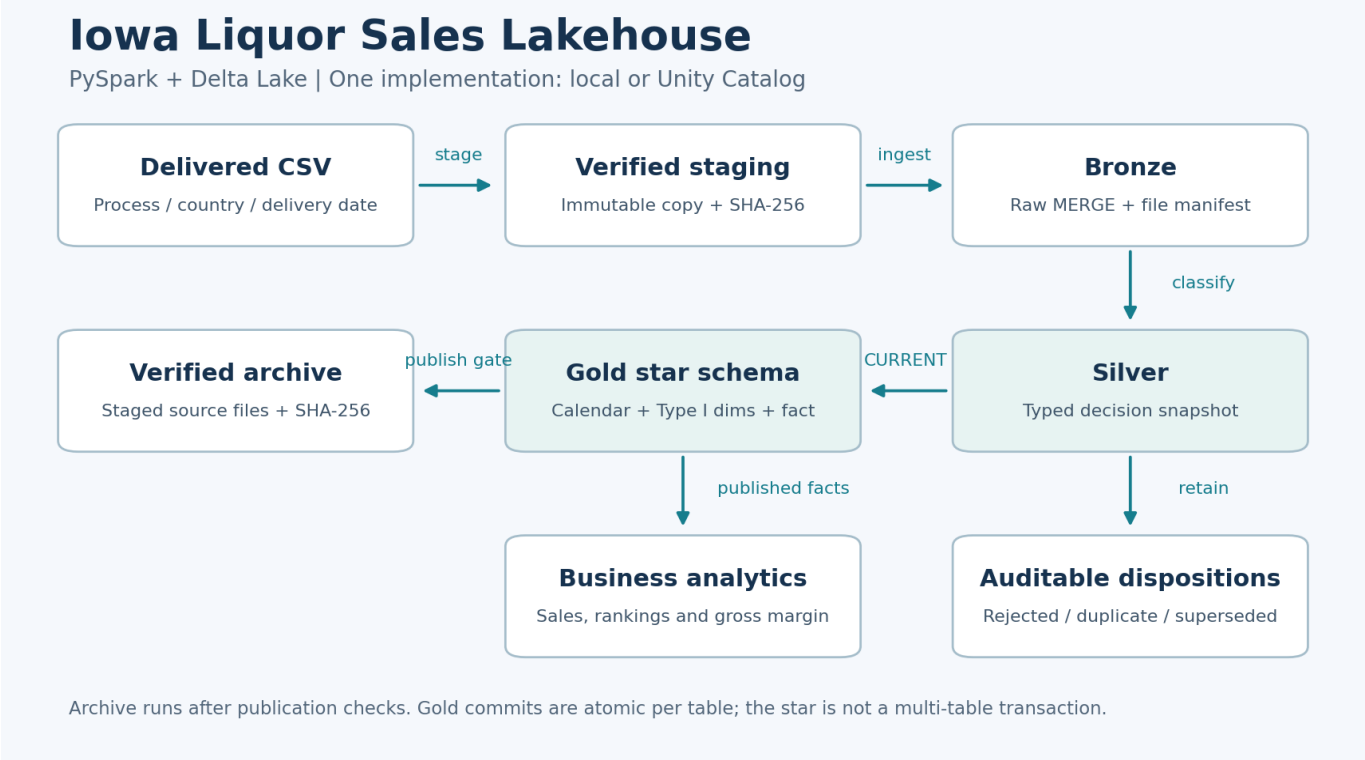
Iowa Liquor Sales Lakehouse

Andres Olguin | Data Engineering Portfolio
PySpark - Spark SQL - Delta Lake - Databricks - Unity Catalog

A reproducible batch pipeline for delivered invoice-line data: immutable raw ingestion, explicit quality decisions, a star schema and verified source archival.



Execution and delivery

One tested Python package serves local Delta execution, interactive notebooks and a versioned Databricks job. The job executes Bronze, Silver, Gold and Archive sequentially, with one active run per job.

Repository: github.com/andresolguin/databricks-medallion-lakehouse-pipeline

Data model and contracts

Gold object	Logical key	Role
dim_time	date_key	Continuous calendar
dim_store	country, store_id	Current store attributes, Type I
dim_item	country, item_no	Current item/category/vendor, Type I
fact_sales	country, invoice_line_no	Measures, dimension keys, source lineage

Fact partitions use the actual sale year/month. Measures use decimal types; ZIP codes remain strings. Keys are checked for uniqueness, non-null values and dimension coverage.

Acceptance rules

Valid sale date; positive store/item identifiers; positive whole pack, bottle-volume and bottle-count fields; positive cost, retail, revenue and sold volumes; retail at least cost. Optional descriptive attributes can be unknown.

Per-line balance tolerances: USD 0.05 for revenue versus bottles x retail; 0.02 L for volume versus bottles x bottle size; 0.02 US gal for liters/gallons conversion. The supported scope is positive sales; returns are quarantined.

Corrections and dispositions

The complete invoice-line key is retained. Latest delivery date determines the current observation. Equivalent latest-day observations deduplicate; different payloads on that date are quarantined as conflicts. Older deliveries arriving late cannot override newer observations.

One Silver Delta table contains CURRENT, REJECTED, DUPLICATE and SUPERSEDED observations. Rejected rows retain reason codes and raw payloads. Occurrence counts preserve physical duplicates, enabling complete row accounting.

Gold reconciles its fact from the complete current Silver snapshot. A later invalid correction removes the corresponding current fact when a nonempty valid snapshot can be published. An entirely invalid snapshot blocks publication.

Reproduce and verify

Local example

```
python -m pip install -e ".[local,dev]"

python examples/generate_inputs.py --root .state/demo

python -m iowa_lakehouse --local-root .state/demo --process-date 20251018

python -m iowa_lakehouse --local-root .state/demo --process-date 20251019

pytest -q
```

Use Python 3.12 and Java 17. Local tests use Spark 4.0.1 with Delta Lake 4.0.0. Initial dependency downloads require internet access. The generated files are entirely synthetic.

Delivery	Current facts	Published sales
20251018	3	USD 270.00
Replay 20251018	3	USD 270.00
20251019	4	USD 366.00

Verification scope

Local automated suite: 33 passed. The tests cover real Delta writes/MERGE, replay, corrections, backfills, conflicting records, field rejection, calendar analytics, partial archival and recovery between raw/manifest commits.

Databricks deployment configuration and commands are included. Workspace authentication, Unity Catalog privileges, serverless execution and target-volume performance require validation in that workspace. Local results do not certify those cloud properties.

Operations and analytics

Verified archival and restart

Content SHA-256 plus country/delivery identifies each file. Raw merges are insert-only by deterministic row ID; receipts record staged/archive paths and status. Retry after an interrupted manifest registration does not reinsert raw rows.

Archive runs after publication verification. It copies staged sources, verifies their hashes, then removes matching inputs and marks receipts complete. A partial archive copy can be repaired from verified Staging. Exceptions fail the task and preserve recoverable inputs.

Analytical outputs

Calendar-based monthly windows; YoY joins to the exact prior-year month; store/category rankings; item gross margins. Annual completeness is supplied as an explicit source watermark. Type I attributes classify historical facts using current dimension values.

Price-volume analysis aggregates one observation per item/month and requires six distinct months with price/quantity variation. Its log-log association is descriptive and does not identify causal demand response or recommend prices.

Design boundaries

Delta atomicity is per table. Gold dimensions and facts have separate commits; Type I changes can be visible before the fact commit. One writer per namespace is required. Silver scans retained Bronze to build its full decision snapshot; this is not a streaming or fully incremental Silver implementation.

Throughput and large-volume capacity are not inferred from the synthetic examples. Measure them on the target runtime and an identified source extract. The project does not make a production-service or cross-table transaction guarantee.

Detailed references

README.md - docs/architecture.md - docs/data_contract.md - docs/runbook.md - docs/validation.md - tests/ - resources/pipeline.job.yml

Domain source: Iowa Data Hub, Iowa Liquor Sales. No third-party source dataset is committed. Project code, documentation and synthetic examples are MIT licensed.